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Research Article

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Posted Date: December 4th, 2023

DOI: <https://doi.org/10.21203/rs.3.rs-3690916/v1>

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Additional Declarations: No competing interests reported.

Optimal Fuzzy Modus Ponens and Tollens Algorithm Based on KSI Principle

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Abstract: L. A. Zadeh has proposed the Compositional rule of inference (CRI) in fuzzy reasoning in 1975. But Zadeh's CRI has some shortcomings that do not satisfy the reductive property. To overcome the shortcomings of the CRI, Compensating Fuzzy Reasoning (CFR) method was proposed by [15]. Its idea is called KSI Principle, which does satisfy the reductive property. In this paper we propose a new optimal approximate reasoning method based on KSI principle. Our new method is based on obtaining fuzzy reasoning results based on moving difference obtained from a given fuzzy premise and fuzzy antecedents, obtaining new fuzzy premises from the moving difference of this reasoning results and the consequent fuzzy set, and then iteratively obtaining the new reasoning results different from the previous reasoning results by fuzzy moving difference obtained from these fuzzy premises and the antecedent fuzzy set until the objective function is satisfied. Proposed method is better in accordance with human thinking than the previous Zadeh's CRI method.

Keywords: Compositional Rule of Inference (CRI), Optimal Fuzzy Modus Ponens (OFMP), Optimal

25 Fuzzy Modus Tollens (OFMT), Optimal Fuzzy Approximate Reasoning (OFAR), Reductive Property, KSI
26 principle

27 **1. Introduction**

28 Research on engineering modeling of human thinking processes has long been carried out. From
29 Socrates' syllogism, modus ponens and modus tollens have been effectively applied to many applications
30 by modeling human thinking. [1] proposed fuzzy set theory, proposed fuzzy modus ponens and fuzzy modus
31 tollens, and discovered fuzzy inference method [1] based on the compositional rule of inference (CRI),
32 which has been widely known by scholars as a powerful mathematical tool for modeling human thinking
33 process, where satisfying reductive property is one of the important properties that inference method should
34 solve.

35 Reductive property refers to the property of inference, in which the knowledge or rule of the form of if-
36 then is accumulated in the human mind and, given new fact information or a new premise, inference results
37 are obtained corresponding to the premise. People often make the following inferences in everyday life [18-
38 22]. For example, when there is a rule called "If the color of this tomato is red, then that tomato is ripe" and
39 the fact information (given premise) is "This tomato is very red", it is said that if "That tomato is very ripe"
40 is obtained as a result of fuzzy inference, this fuzzy system satisfies reductive property completely. People
41 do not know scientifically clearly how they think and calculate these results, but this result is natural.
42 However, fuzzy inference is a mathematical computational tool constructed by imitating human thinking,
43 and the results are different depending on what operation is used. For example, the compositional rule (CRI)
44 of Zadeh's inference uses min-max operations, whose inference results are not satisfactory for reductive
45 property.

46 In a SISO (single Input Single Output) fuzzy system with one rule, reductive property is probably
47 satisfied, but in the case of n rules the reductive property is probably not satisfied. Moreover, in practice,

several or tens of fuzzy rules are used, so the problem of satisfying reductive property becomes serious in MISO (Multiple Input Single Output) fuzzy systems or MIMO (Multiple Input Multiple Output) fuzzy systems with n rules. Therefore, the study of fuzzy inference methods satisfying reductive property is an important research index in many fields where artificial intelligence is applied, including economy and defense.

Numerous researchers have been published to solve the reductive property problem of fuzzy inference. To overcome the disadvantages of Zadeh's method, tens of fuzzy inference methods have been proposed, such as Triple Implication Principle (TIP), Similarity-based Inference (SI), Interpolation based Inference (II), and Quintuple Implication Principle (QIP) [2-5, 11-14, 16, 18, 19]. However, it can be seen that none of these methods are essential improvements as alternative methods based on CRI proposed by Zadeh. Although many scientific practices based on Zadeh's inference method have proved a lot of effectiveness over the past 50 years, the Zadeh's method still has the disadvantage that its meaning is linguistically imprecise and does not satisfy reductive property. In 1987, in [21], the author pointed out that principle other ideas and methodologies should be proposed to solve this problem, since inference by the compositional rule of inference by Zadeh does not satisfy reductive property. On the other hand, in [6,7], a new idea of fuzzy inference based on optimization techniques has been proposed. With reductive property robustness is an important property to be examined in practical applications of fuzzy inference, and so far a lot of studies have been reported. The reductive property of fuzzy inference has been studied in many literatures [15-24], and the topic of robustness (Robustness) has been studied in [8,9] in the setting of δ -equivalence of fuzzy sets and extended in [10].

The idea of the optimal fuzzy inference method based on Zadeh's CRI can be described as follows [6,7,13].

- 1) Assume that a fuzzy relation (matrix) R is obtained from expert experience (fuzzy rule) or data.

We must then trust R and treat it as the basis for evaluating the quality of the fuzzy inference method.

- 2) Given a fuzzy premise A^* , the fuzzy inference method will generate the corresponding fuzzy conclusion B^* . If the new matrix R' in the inference process is close to R with certain sensitivity, then the fuzzy inference method can be said to be optimal. Here, $R' = A \rightarrow B$, which means that if A then B .

Optimal fuzzy inference is based on the optimization of the objective function in the acquisition of inference results. In [13], the author mentioned that for a fuzzy rule, it is not appropriate to describe "if A then B " from the view of optimal fuzzy inference, instead the fuzzy rule should be described differently as "if A then B , for goal J_{op} ". Here, J_{op} is an objective function. Given a new given fuzzy premise $A_i^* = [a_1^*, a_2^*, \dots, a_m^*]$, $a_i^* = \mu_{A_i^*}(x)$, the reasoning result $B^* = [b_1^*, b_2^*, \dots, b_n^*]$, $b_j^* = \mu_{B_j^*}(y)$ is obtained by a certain inference method or inference algorithm. Based on the idea mentioned above, the authors [13] discussed two inference purposes/limitations. One is the perturbation of the rule and the other is the perturbation of the fuzzy consequent. On the other hand, [15] proposed a new inference principle based on compensation operation to obtain a new inference result by reflecting it in the consequent part, and demonstrated its validity and effectiveness by considering that there is a moving and transformation relationship between a given premise as input information and the antecedent fuzzy set of fuzzy rules without depending on the traditional compositional rules of inference of Zadeh. In [16], authors proposed a method for obtaining new inference results by calculating the moving distance between the antecedent and the premise by the Euclidean distance measure, and theoretically and experimentally demonstrate that higher reductive property than reductive property of previous methods is obtained and better suited to human thinking.

L. A. Zadeh has proposed the Compositional rule of inference (CRI) in fuzzy reasoning in [1]. But Zadeh's CRI has some shortcomings that do not satisfy the reductive property. To overcome the

shortcomings of the CRI, Compensating Fuzzy Reasoning (CFR) method was proposed by S. I. Kwak et al. in [15]. Its idea is called KSI Principle, which does satisfy the reductive property. In this paper, we describe an optimal fuzzy approximate reasoning algorithm based on distance measure that can open a new research direction in the field of fuzzy inference for modeling engineering human thinking process. In this paper, we propose a new fuzzy approximate reasoning algorithm using a distance measure based on [15,16,17] and an objective function based on [13] without using fuzzy relation [1], and discuss its reductive property, unlike the optimal fuzzy approximate reasoning proposed based on fuzzy relation and objective function in the previous literature. This paper is organized as follows. In Section 2, we consider the previous optimal fuzzy reasoning methods. In Section 3, a new optimal fuzzy reasoning (OFAR) algorithm based on distance measure is proposed for SISO and discrete MISO fuzzy systems and its properties are described. Section 4 presents the simulation and results analysis. Finally, conclusions are discussed in Section 5.

2. Related Works

According to [16,21,26], the fuzzy inference scheme in the discrete SISO fuzzy system can be written as (1).

$$\begin{aligned} &\text{Rule; if } x \text{ is } A \text{ then } y \text{ is } B, \text{ Premise: } x \text{ is } A^*, \text{ Conclusion: } y \text{ is } B^*, \text{ for FMP} \\ &\text{Rule; if } y \text{ is } \overline{B} \text{ then } x \text{ is } \overline{A}, \text{ Premise: } y \text{ is } B^*, \text{ Conclusion: } x \text{ is } A^*, \text{ for FMT} \end{aligned} \quad (1)$$

Here, $A \in F(X)$, $A^* \in F(X)$ are fuzzy sets defined on the whole set X , $B \in F(Y)$, $B^* \in F(Y)$ are fuzzy sets defined on the whole set Y considered, and $\overline{A} = 1 - A$, $\overline{B} = 1 - B$. In 2020 in [16], based on [22,27], the reductive property criteria of fuzzy inference is proposed as in Table 1.

Table 1. Reductive property criteria for FMP and FMT based on [16, 22, 23]

FMP; if x is A then y is B		
case	x is A^* (premise)	Reductive property criteria $RPCF_{FMP}$ (%)

1	$A^* = A$	$RPCF_{FMP} = (1 - \sum_{k=1}^{\Theta} b_{kl}^* - b_k / \Theta) \times 100$
2	$A^* = A^2$	$RPCF_{FMP} = (1 - \sum_{k=1}^{\Theta} b_{kl}^* - b_k^2 / \Theta) \times 100$
3	$A^* = A^{1/2}$	$RPCF_{FMP} = (1 - \sum_{k=1}^{\Theta} b_{kl}^* - b_k^{1/2} / \Theta) \times 100$
4	$A^* = 1 - A$	$RPCF_{FMP} = (1 - \sum_{k=1}^{\Theta} b_{kl}^* (1 - b_k) / \Theta) \times 100$
5	$A^* = s.t. A$	$RPCF_{FMP} = (1 - \sum_{k=1}^{\Theta} b_{kl}^* - s.t. b_k / \Theta) \times 100$
FMT; if y is $\overline{B}$ then x is $\overline{A}$		
case	y is B^* (premise)	Reductive property criteria $RPCF_{FMT}$ (%)
6	$B^* = 1 - B$	$RPCF_{FMT} = (1 - \sum_{k=1}^{\Theta} a_{kl}^* - (1 - a_k) / \Theta) \times 100$
7	$B^* = 1 - B^2$	$RPCF_{FMT} = (1 - \sum_{k=1}^{\Theta} a_{kl}^* - (1 - a_k)^2 / \Theta) \times 100$
8	$B^* = 1 - B^{1/2}$	$RPCF_{FMT} = (1 - \sum_{k=1}^{\Theta} a_{kl}^* - (1 - a_k)^{1/2} / \Theta) \times 100$
9	$B^* = B$	$RPCF_{FMT} = (1 - \sum_{k=1}^{\Theta} a_{kl}^* - a_k / \Theta) \times 100$
10	$B^* = s.t. B$	$RPCF_{FMT} = (1 - \sum_{k=1}^{\Theta} a_{kl}^* - s.t. a_k / \Theta) \times 100$

113 In Table 1, s.t means "slightly tilted of" defined in [26]. In Table 1, Case 5 and Case 10, as proposed in
 114 [26], are almost similar to Case 1 and Case 9, respectively. For SISO systems with one rule, the results of
 115 fuzzy inference and reductive property evaluation of the previous DMM method [16] in class 1 are shown
 116 in Table 2.

117 **Table 2.** Result and reductive property of FMP-DM and FMT-DM in $P(+1,-1)$ form [16]

Reasoning result and reductive property of FMP-DM			
Case	Premise $A^*(x)$	Reasoning result $B^*(y)$	Reductive property
1	[1, 0.3, 0, 0, 0]	[0, 0, 0, 0.3, 1]	100 %

2	[1, 0.09, 0, 0, 0]	[0.158, 0, 0.158, 0.41, 1]	87.26 %
3	[1, 0.548, 0, 0, 0]	[0, 0, 0, 0.3, 1]	95.05 %
4	[0, 0.7, 1, 1, 1]	[0, 1, 1, 0.84, 0.45]	68.25 %
<i>RPCF FMP – DM – p(+1,-1) form</i>			87.64 %
Reasoning result and reductive property of FMT-DM			
Case	Premise $B^*(y)$	Reasoning result $A^*(x)$	Reductive property
6	[1, 1, 1, 0.7, 0]	[0, 0.7, 1, 1, 1]	100.0%
7	[1, 1, 1, 0.91, 0]	[0, 0.7, 1, 1, 1]	95.80 %
8	[1, 1, 1, 0.452, 0]	[0, 0.7 1, 0.778, 1]	90.61 %
9	[0, 0, 0, 0.3, 1]	[0.55, 0.16, 0, 0, 1]	68.25 %
<i>RPCF FMT – DM – p(+1,-1) form</i>			88.67 %

118 According to [1], the CRI solution of FMP and FMT is as follows.

119 FMP-CRI; $B^*(y) = \bigvee_{x \in U} (A^*(x) \otimes (A(x) \rightarrow B(y)))$ (2)

120 FMT-CRI; $A^*(x) = \bigvee_{y \in V} (B^*(y) \otimes (A(x) \rightarrow B(y)))$ (3)

121 The inference results and reductive properties of the previous FMP-CRI and FMT-CRI by the implication
122 of Łukasiewicz in the discrete SISO fuzzy system are shown in Table 3.

123 **Table 3.** Reductive property of FMP-CRI and FMT-CRI by the implication of Łukasiewicz in SISO
124 fuzzy system

FMP-CRI-Conclusion $B^*(y)$ and Reductive Property		
Case	Premise $A^*(x)$	FMP- CRI –Łukasiewicz

1	[1, 0.3, 0, 0, 0]	[0, 0, 0, 0.3, 1]	100 %
2	[1, 0.09, 0, 0, 0]	[0, 0, 0, 0.3, 1]	95.8 %
3	[1, 0.548, 0, 0, 0]	[0.248, 0.248, 0.248, 0.541]	85.14 %
4	[0, 0.7, 1, 1, 1]	[1, 1, 1, 1, 1]	74.00 %
$RPCF_{FMP-CRI-I}$			88.73 %
FMT-CRI-Conclusion $A^*(x)$ and Reductive Property			
Case	Premise $B^*(y)$	FMT-CRI-Łukasiewicz	
6	[1, 1, 1, 0.7, 0]	[1, 1, 1, 1, 1]	74.00 %
7	[1, 1, 1, 0.91, 0]	[1, 1, 1, 1, 1]	78.20 %
8	[1, 1, 1, 0.452, 0]	[1, 1, 1, 1, 1]	69.05 %
9	[0, 0, 0, 0.3, 1]	[1, 1, 1, 1, 1]	26.00 %
$RPCF_{FMT-CRI-I}$			61.81 %

125 The reductive property comparison results of CRI, TIP, QIP, AARS and DMM presented in [16] for SISO
126 fuzzy systems are shown in Table 4.

127 **Table 4.** Comparisons of CRI, TIP, QIP, AARS and DMM reported in [16]

No	Fuzzy Reasoning Method		$RPCF_{FR}$
1	CRI [1]	Gödel; G	77.131 %
2	TIP [24]	Gödel; G	68.570 %
3	QIP [23]	Gödel; G	61.950 %
4	AARS [25]	<i>more or less form</i>	57.121 %
5	DMM [16]	$P_{(+1,-1)}$ form	90.310 %

129 In this section, a new reductive property estimation algorithm of optimal fuzzy approximate reasoning
130 based on KSI Principle is formulated for fuzzy modus ponens and fuzzy modus tollens in discrete MISO
131 fuzzy systems.

132 3.1 New ideas

133 The idea of this paper is based on first obtaining reasoning results based on fuzzy distance measures
134 obtained from the given premise and antecedent fuzzy set, obtaining new fuzzy premises from these
135 reasoning results and the moving distance of this reasoning result and the consequent fuzzy set, and
136 repeatedly obtaining new reasoning results other than the previous reasoning results until the objective
137 function satisfies the desired performance indices by fuzzy moving distance measure obtained from this
138 fuzzy premise and antecedent fuzzy set. In [13], authors have improved Zadeh's CRI method by setting the
139 objective function by the fuzzy relation matrix R and applying the optimization method to inference. The
140 basic idea of optimal fuzzy reasoning based on the distance measure proposed in this paper is as follows.

141 The rule base is not represented as a fuzzy relation matrix R in traditional fuzzy inference, but consists
142 of a pair of a distance measure DM_{ant} obtained from the antecedent of the fuzzy rule and a given premise
143 and distance measures DM_{cons} obtained from the consequent of the fuzzy rule and the conclusion obtained
144 by the inference method. The former is called the antecedent distance measure, the latter is called the
145 consequent distance measure. In fuzzy reasoning by distance measure, assume that the rule base is
146 represented by this distance measure $DM = f(DM_{ant}, DM_{cons})$. The expert then uses this distance measure
147 $DM = f(DM_{ant}, DM_{cons})$ to evaluate the reasoning.

148 First, the inference gets the antecedent distance measure DM_{ant} from the given fuzzy premise A^* and the
149 antecedent A , and obtain a new reasoning result B^* by reflecting this into the consequent part B . Then we
150 obtain the consequent distance measure DM_{cons} between the consequent B and the reasoning result B^* . We

151 obtain a new premise A^* by mapping this consequent distance measure DM_{cons} to the antecedent A . During
 152 the inference, given the fuzzy premise A^* , the corresponding conclusion B^* will be generated through some
 153 certain inference method (through DM_{ant}). Then this pair of A^* and B^* forms a new distance measure DM'_{cons} ,
 154 which is different from the original DM_{cons} for most fuzzy reasoning models (FRMs). In the proposed
 155 optimal fuzzy reasoning method, the feedback introduced during the reasoning process will search for the
 156 best optimal B^* from the conclusion region to minimize the "reasoning error" defined by the objective
 157 function (4).

$$158 \quad J(DM_{cons}, DM'_{cons}) = \|DM_{cons} - DM'_{cons}\| \quad (4)$$

159 For the previous inference model, the objective function is usually $J(DM_{cons}, DM'_{cons}) \neq 0$. For the optimal
 160 reasoning process based on distance measure, the reasoning result B^* is generally difficult to obtain
 161 analytical results, which are obtained by a constant preconditioning method. Its basic idea is to continuously
 162 adjust the reasoning result B^* according to the change in (4). About the objective function $J(DM_{cons}, DM'_{cons})$,
 163 the evaluation criteria given by the designer, the computation is satisfied is stopped. It is because evaluating
 164 analytically reasoning result B^* is very complex and difficult to implement. From the above idea, it is easy
 165 to see that the distance measure-based optimal fuzzy reasoning model (OFRM) can treat general fuzzy
 166 inference as an implementation of one optimization algorithm instead of a logical inference procedure. In
 167 general, since the inner loop used in optimal fuzzy approximate reasoning is within the feedback loop of
 168 the whole fuzzy system under consideration, the effect of disturbance such as perturbation of the reasoning
 169 premise due to noise of inputs in the fuzzy reasoning process can be greatly reduced [6,7,13].

170 3.2 Novel optimal fuzzy approximate reasoning algorithm for MISO fuzzy system

171 In this subsection, we consider both the two types of $P(+1,0,-1)_{form}$ and $P(+1,-1)_{form}$ as sign vectors to
 172 take into account the effect of the sign vector on the reductive property in the case that discrete MISO fuzzy

173 systems consist of n rules.

174 The optimal fuzzy modus ponens scheme of discrete MISO fuzzy system composed of n rules is as
175 follows.

176 $R_i : \text{if } x_1 \text{ is } A_{i1}(x) \text{ and } \cdots x_j \text{ is } A_{ij}(x) \cdots \text{and } \cdots x_m \text{ is } A_{im}(x) \text{ then } y_i \text{ is } B_i(y)$

177 **Premise;** $x_1 \text{ is } A_1^*(x) \text{ and } x_2 \text{ is } A_2^*(x) \text{ and } \cdots x_j \text{ is } A_j^*(x) \cdots \text{and } \cdots x_m \text{ is } A_m^*(x)$

178 **Conclusion:** $y \text{ is } B^*(y), \text{ for goal } J_{\text{optimal}} \text{FMP}$ (5)

179 Where $R_i (i=1, 2, \dots, n)$ denotes the fuzzy rule and $x_j \in X, j=1, 2, \dots, m$ denotes the input variable of the
180 object. And $y \in Y$ represents the output variable of the object, $A_j^*(x) \subset X, A_{ij}(x) \subset X, B_i(y) \subset Y,$
181 $B^*(y) \subset Y, (i=1, 2, \dots, n; j=1, 2, \dots, m)$ is the fuzzy set of input information and antecedent, consequent and
182 reasoning results, respectively, and X, Y is the whole set where the object system is defined. In (5), the fuzzy
183 set $a_{ij}^k, a_{jl}^{k^*}, k=1, 2, \dots, r$ of the antecedent is given by the fuzzy sets of the antecedent, respectively, and the
184 fuzzy set vectors of the given whole are as follows.

185 $A_{ij} = [a_{ij}^1, a_{ij}^2, \dots, a_{ij}^k, \dots, a_{ij}^r]$ (6)

186 $A_{jl}^* = [a_{jl}^{1^*}, a_{jl}^{2^*}, \dots, a_{jl}^{k^*}, \dots, a_{jl}^{r^*}]$ (7)

187 For discrete MISO fuzzy systems composed of n rules, the optimal fuzzy modus tollens scheme is as
188 follows.

189 $R_i : \text{if } y_i \text{ is } \overline{B_i(y)} \text{ then } x_1 \text{ is } \overline{A_{i1}(x)} \text{ and } \cdots x_j \text{ is } \overline{A_{ij}(x)} \cdots \text{and } \cdots x_m \text{ is } \overline{A_{im}(x)}$

190 **Premise;** $y \text{ is } B^*(y)$

191 **Conclusion:** $x_1 \text{ is } A_1^*(x) \text{ and } x_2 \text{ is } A_2^*(x) \text{ and } \cdots x_j \text{ is } A_j^*(x) \cdots \text{and } \cdots x_m \text{ is } A_m^*(x), \text{ for goal } J_{\text{optimal}} \text{FMT}$ (8)

192 The antecedent fuzzy set vector $\overline{B_i(y)}$ of the consisting of antecedent fuzzy set $b_i^k, b_j^{k^*}, k=1, 2, \dots, r$ and
193 fuzzy set vector $B_i^*(y)$ of the given premise are as follows:

194 $\overline{B_i} = [\overline{b_i^1}, \overline{b_i^2}, \dots, \overline{b_i^k}, \dots, \overline{b_i^r}], B_i^* = [b_i^{1^*}, b_i^{2^*}, \dots, b_i^{k^*}, \dots, b_i^{r^*}]$ (9)

The new proposed optimal fuzzy reasoning algorithm for the discrete MISO fuzzy system denoted by (8) and (9) is as follows.

Optimal Fuzzy Approximate Reasoning Algorithm

Stage 1: Let the number of iterations as $t = 1$.

Stage 2: For discrete MISO fuzzy systems, the process of extending the antecedent discrete fuzzy set vector $A_{ij}(\text{resp. } \bar{B}_i)$, ($i = 1, 2, \dots, n$, $j = 1, 2, \dots, m$), the premise discrete fuzzy set vector $A_i^*(\text{resp. } B^*)$ and the consequent fuzzy set $B_i(\text{resp. } \bar{A}_{ij})$ by a least common multiple is denoted by $\xrightarrow{LCM}$, and the extended fuzzy row vectors are calculated as follows.

Stage 2-1: Using a least common multiplier of extended vectors of FMP in discrete MISO fuzzy systems, we extend the fuzzy set vectors of fuzzy rule and the premise fuzzy set vectors as shown in Eq. (6) - (7) above.

$$A_{ij} \xrightarrow{LCM} \tilde{A}_{ij} = \left[\frac{a_{ij}^1}{x_1}, \dots, \frac{a_{ij}^k}{x_r}, \dots, \frac{a_{ij}^\Theta}{x_\Theta} \right] \quad (10)$$

$$A^* \xrightarrow{LCM} \tilde{A}_{ij}^* = \left[\frac{a_{ij}^{1*}}{x_1}, \dots, \frac{a_{ij}^{k*}}{x_r}, \dots, \frac{a_{ij}^{\Theta*}}{x_\Theta} \right] \quad (11)$$

$$B \xrightarrow{LCM} \tilde{B} = \left[\frac{b_1}{y_1}, \dots, \frac{b_r}{y_r}, \dots, \frac{b_\Theta}{y_\Theta} \right] \quad (12)$$

Stage 2-2: Extending the premise fuzzy set vectors and fuzzy set vectors of the fuzzy rule shown in Eq. (9) above by the least common multiplier of extension vector of FMT in the discrete MISO fuzzy system is as follows.

$$\bar{B}_i \xrightarrow{LCM} \tilde{\bar{B}}_i = \left[\frac{\bar{b}_i^1}{y_1}, \dots, \frac{\bar{b}_i^k}{y_k}, \dots, \frac{\bar{b}_i^\Theta}{y_\Theta} \right] \quad (13)$$

$$B_i \xrightarrow{LCM} \tilde{B}_i^* = \left[\frac{\bar{b}_i^{1*}}{y_1}, \dots, \frac{\bar{b}_i^{k*}}{y_k}, \dots, \frac{\bar{b}_i^{\Theta*}}{y_\Theta} \right] \quad (14)$$

$$A_j \xrightarrow{LCM} \tilde{A}_j^* = [\frac{a_j^{1^*}}{x_1}, \dots, \frac{a_j^{k^*}}{x_r}, \dots, \frac{a_j^{\Theta^*}}{x_\Theta}] \quad (15)$$

Stage 3: In discrete MISO fuzzy system, the extended distance measure $EDM_{ijl}(\tilde{A}_{ij}^*, \tilde{A}_{ij})_{i\Theta}$, $EDM_{jl}(\tilde{B}_l^*, \tilde{B}_l)_{i\Theta}$ of the extended antecedent vector of fuzzy rule and the premise vector $\tilde{A}_{ij}, \tilde{A}_{ij}^*, j=1,2,\dots,m$, (*resp.* $\tilde{B}_l, \tilde{B}_l^*$, $i=1,2,\dots,n, l=6,7,8,9$) are calculated as follows.

$$\begin{aligned} EDM_{ijl}(\tilde{A}_{ij}^*, \tilde{A}_{ij})_{i\Theta} &= [\frac{1}{\Theta} \sum_{k=1}^{\Theta} [a_{ijl}^{k^*} - a_{ijl}^k]^2]^{1/2} \text{ for OFMP} \\ EDM_{jl}(\tilde{B}_l^*, \tilde{B}_l)_{i\Theta} &= [\frac{1}{\Theta} \sum_{k=1}^{\Theta} [b_l^{k^*} - \bar{b}_l^k]^2]^{1/2} \text{ for OFMT} \end{aligned} \quad (16)$$

The words of optimal fuzzy modus ponens and optimal fuzzy modus tollens are respectively abbreviated as OFMP and OFMT.

Stage 4: In the discrete MISO fuzzy system, the sign vectors $\tilde{P}_l$ by the difference between the given premise and the antecedent $dif_{mkl}, (k = \overline{1, lcm(u, v)}, l=1,2,\dots,s)$ are divided into two sign vectors $P(+1,0,-1)$ and $P(+1,-1)$, respectively, and calculated as follows:

$$\tilde{P}_{ml} = [\tilde{P}_{m1l}, \tilde{P}_{m2l}, \dots, \tilde{P}_{mkl}, \dots], \text{ for OFMP, OFMT} \quad (17)$$

$$P(+1,0,-1) \text{ form; } P_{mkl} = \text{sign}(dif_{mkl}) = \begin{cases} +1, & dif_{mkl} > 0 \\ 0, & dif_{mkl} = 0 \\ -1, & dif_{mkl} < 0 \end{cases} \quad (18)$$

$$P(+1,-1) \text{ form; } P_{mkl} = \text{sign}(dif_{mkl}) = \begin{cases} +1, & dif_{mkl} \geq 0 \\ -1, & dif_{mkl} < 0 \end{cases} \quad (19)$$

$$dif_{mkl} = \begin{cases} a_{mkl}^* - a_{mkl}, & \text{for OFMP} \\ b_{mkl}^* - \bar{b}_{mkl}, & \text{for OFMT} \end{cases} \quad (20)$$

Stage 5: Since the fuzzy distance measure $EDM_{ijl}(\tilde{A}_{ij}^*, \tilde{A}_{ij})_{i\Theta}$, $EDM_{jl}(\tilde{B}_l^*, \tilde{B}_l)_{i\Theta}$ with index $\Theta = lcm(u, v)$ in the discrete MISO fuzzy system is a scalar quantity, the vectorized distance measure $\tilde{C}_l$ is calculated as follows:

$$\tilde{C}_l = \begin{cases} \frac{1}{m} (\sum_{j=1}^m EDM_{ijl}(\tilde{A}_{ij}^*, \tilde{A}_{ij})_{i\Theta} \times \tilde{P}_{jl}), & \text{for OFMP} \\ EDM_{jl}(\tilde{B}_l^*, \tilde{B}_l)_{i\Theta} \times \tilde{P}_l, & \text{for OFMT} \end{cases} \quad (21)$$

Stage 6: For OFMP and OFMT in the discrete MISO fuzzy system, the quasi-quasi-reasoning result

233 $\tilde{B}_l^{**}, \tilde{A}_{jl}^{**}, l=1,2,\dots,s, i=1,2,\dots,m$ is obtained as follows.

$$234 \quad \tilde{B}_l^{**} = \begin{cases} \tilde{B}_l + \tilde{C}_l, & \text{if Case 1, 2, and 3} \\ 1 - \tilde{B}_l + \tilde{C}_l, & \text{if Case 4, for OFMP,} \\ s.t. \tilde{B}_l + \tilde{C}_l & \text{if Case 5} \end{cases} \quad \tilde{A}_{jl}^{**} = \begin{cases} \tilde{A}_{jl} + \tilde{C}_i, & \text{if Case 6,7, and 8} \\ 1 - \tilde{A}_{jl} + \tilde{C}_i, & \text{if Case 9, for OFMT} \\ s.t. \tilde{A}_{jl} + \tilde{C}_i & \text{if Case 10} \end{cases} \quad (22)$$

235 **Stage 7:** In the discrete MISO fuzzy system, the process of reducing the original dimensions from

236 $\tilde{B}_l^{**}, \tilde{A}_{jl}^{**}$ to $\tilde{B}_l^{**}, \tilde{A}_{jl}^{**}$ is called the inverse minimum co-multiplication method, and the resulting quasi-fuzzy

237 $B_l^{**}, A_{jl}^{**}, j=1,2,\dots,m, l=1,2,\dots,s$ approximate inference is chosen as follows.

$$238 \quad \begin{aligned} \tilde{B}_l^{**} &\xrightarrow{LCM^{-1}} B_l^{**}, i.e., [\tilde{b}_{lk}^{**}]_{\Theta \times 1} \xrightarrow{LCM^{-1}} [b_{lk}^{**}]_{v \times 1}, \text{ for OFMP} \\ \tilde{A}_{jl}^{**} &\xrightarrow{LCM^{-1}} A_{jl}^{**}, i.e., [\tilde{a}_{jk}^{**}]_{\Theta \times 1} \xrightarrow{LCM^{-1}} [a_{jk}^{**}]_{u \times 1}, \text{ for OFMT} \end{aligned} \quad (23)$$

239 Here the index is $\Theta = lcm(u, v) = u \cdot m_1 = v \cdot m_2$ and m_1, m_2 are integers.

240 **Stage 8:** In discrete MISO system, from the quasi-fuzzy approximate reasoning results

241 $B_l^{**}, A_{jl}^{**} (j=1,2,\dots,m)$ the fuzzy approximate reasoning results B_{jl}^{*}, A_{jl}^{*} normalized the values greater than 1

242 and less than 0 to values between 1 and 0, respectively, are obtained as follows.

$$243 \quad \begin{aligned} B_l^{*} &= (B_l^{**} - \eta_l) / (\xi_l - \eta_l), \quad j=1,2,\dots,m, l=1,2,3,4, \text{ for OFMP} \\ A_{jl}^{*} &= (A_{jl}^{**} - \eta_{jl}) / (\xi_{jl} - \eta_{jl}), \quad j=1,2,\dots,m, l=6,7,8,9, \text{ for OFMT} \end{aligned} \quad (24)$$

244 Here, the maximum matrix ξ_{jl} and minimum η_{jl} of the quasi-fuzzy approximate reasoning results

245 B_l^{**}, A_{jl}^{**} for OFMP and OFMT are as follows, respectively.

$$246 \quad \begin{aligned} \xi_l &= \max_{1 \leq k \leq r} B_l^{**} \quad l=1,2,3,4, & \text{for OFMP} \\ \eta_l &= \min_{1 \leq k \leq r} B_l^{**} \quad l=1,2,3,4, & \text{for OFMP} \\ \xi_{jl} &= \max_{1 \leq k \leq r} A_{jl}^{**} \quad j=1,2,\dots,m, l=1,2,3,4, & \text{for OFMT} \\ \eta_{jl} &= \min_{1 \leq k \leq r} A_{jl}^{**} \quad j=1,2,\dots,m, l=1,2,3,4, & \text{for OFMT} \end{aligned} \quad (25)$$

247 **Stage 9:** Distance measure $EDM(B_{iml}^{*}, B_l)_{\Theta}, EDM(A_{il}^{*}, \bar{A}_i)_{\Theta}$ of the consequent of fuzzy rules and fuzzy

248 approximation reasoning results in the discrete MISO fuzzy system is called the optimal objective function

249 of OFMP and OFMT, denoted by $J_{i(B_l^{*}, B)}^{Optimal}, J_{i(A_l^{*}, \bar{A})}^{Optimal}$, respectively, and calculated as follows.

$$J_{i-Optimal} = \begin{cases} J_{i-(B_i^*, B)}^{Optimal} = [\frac{1}{\Theta} \sum_{r=1}^{\Theta} [b_{rl}^* - b_r]^2]^{1/2}, & \text{for OFMP} \\ \frac{1}{m} \sum_{j=1}^m J_{i-(A_{jl}^*, \bar{A}_{jl})}^{Optimal} = \frac{1}{m} [\frac{1}{\Theta} \sum_{r=1}^{\Theta} [a_{jrl}^* - \bar{a}_{jr}]^2]^{1/2}, & \text{for OFMT} \end{cases} \quad (26)$$

Stage 10: For OFMP and OFMT of the discrete MISO fuzzy system, if m optimal objective functions (distance measure of the consequent) satisfy the following conditional expression (27), go to Stage 14 or go to Stage 11.

$$J_{i-Optimal} = \begin{cases} J_{i-(B_i^*, B)}^{Optimal} \leq \varepsilon, & \text{for OFMP} \\ \frac{1}{m} \sum_{j=1}^m J_{i-(A_{jl}^*, \bar{A}_{jl})}^{Optimal} \leq \varepsilon, & \text{for OFMT} \end{cases} \quad (27)$$

Here ε is a threshold set according to the characteristics of the object considered by the designer to obtain the optimal fuzzy approximate reasoning result.

Stage 11: Set the of iterations as $t \leftarrow t + 1$.

Stage 12: For OFMP and OFMT of discrete MISO fuzzy system, the new premise vector $\bar{A}_{jl}^*, l=1,2,3,4$ for FMP, $\bar{B}_l^*, l=6,7,8,9$ for FMT from the distance measure of the consequent and antecedent are generated as follows.

$$\begin{aligned} \bar{A}_{jl}^* &= A_{jl} + EDM(B_l^*, B)_{\Theta}, j=1,2,\dots,m, \quad \text{for OFMP} \\ \bar{B}_l^* &= \bar{B} + \frac{1}{m} \sum_{j=1}^m EDM(A_{jl}^*, \bar{A}_{jl})_{\Theta}, \quad \text{for OFMT} \end{aligned} \quad (28)$$

Stage 13: Go to Stage 3.

Stage 14: From the l partial reasoning results $B_l^*(y)$ for OFMP and OFMT of discrete MISO fuzzy system (from $m \times l$ partial reasoning results $A_{jl}^*(y)$), final l ($m \times l$) are calculated as follows:

$$B_l^*(y) = \arg \min_{EDM_l} \{B_{il}^*(y), B_{i2}^*(y), \dots, B_{is}^*(y)\} \quad (29)$$

$$A_{jl}^* = \arg \min_{EDM_l} \{A_{jml}^*(y), A_{jm2}^*(y), \dots, A_{jms}^*(y)\} \quad (30)$$

Stage 15: The final optimal fuzzy approximate reasoning result $B^*, A_j^*, j=1,2,\dots,m$ according to the premises repeatedly obtained for OFMP and OFMT of the discrete MISO fuzzy system is obtained as

269 follows, respectively.

$$270 \quad \begin{aligned} B^* &= \{B_l^*\}, l = 1, 2, 3, 4, & \text{for } OFMP \\ A_j^* &= \{A_{jl}^*\}, l = 6, 7, 8, 9 & \text{for } OFMT \end{aligned} \quad (31)$$

271 **Stage 16:** For the discrete MISO fuzzy system, the overall reductive property $RPCF_{OFMP}^l$ of the optimal
 272 fuzzy approximate reasoning algorithm is obtained as follows by using the reductive property criteria
 273 $RPCF_{Optimal}(\%)$ for OFMP and $RPCF_{OFMT}^l$ for OFMT according to Table 4.

$$274 \quad \begin{aligned} RPCF_{Optimal} &= \frac{1}{2}(RPCF_{OFMP} + RPCF_{OFMT}) \\ RPCF_{OFMP} &= \frac{1}{4} \sum_{l=1}^4 RPCF_{OFMP}^l \\ RPCF_{OFMT} &= \frac{1}{4} \sum_{l=6}^9 RPCF_{OFMT}^l \end{aligned} \quad (32)$$

275 **Stage 17:** Complete algorithm computation of discrete MISO fuzzy system.

276 The following subsection describes the properties of the proposed optimal fuzzy approximate reasoning
 277 algorithm.

278 3.3 Properties of the proposed method

279 **Theorem 3.1** Optimal fuzzy modus ponens OFMP-EDM by proposed method OAFR approximate
 280 inference result satisfies reductive property by 100% and is completely in accordance with human
 281 thinking, when the premise is given by (33) for the discrete SISO fuzzy system .

$$282 \quad A^* = A, \text{ for } OFMP \quad (33)$$

283 **Theorem 3.2** For a SISO fuzzy system, let the premise be given by (34).

$$284 \quad A^* = A^2, A^{1/2}, \bar{A}, \text{ for } OFMP, B^* = \bar{B}^2, \bar{B}^{1/2}, B, \text{ for } OFMT \quad (34)$$

285 Then, the approximate reasoning result of the optimal fuzzy modus tollens OFMT-EDM by the
 286 proposed method OAFR satisfies the reductive property by 100% and is completely in accordance with

human thinking.

Theorem 3.3 For a discrete MISO fuzzy system represented by Eq. (34), the premise is given by (35).

$$A_1^*(x) = A_{i1}(x) \text{ and } A_2^* = A_{i2}(x) \text{ and } \cdots A_j^*(x) = A_{ij}(x) \cdots \text{ and } \cdots A_m^*(x) = A_{im}(x) \quad (35)$$

Approximate reasoning results by OFMP-EDM are obtained as

$$B^*(y) = B_i(y) \quad (36)$$

and satisfy the reductive property by 100%, and thus fully in accordance with human thinking.

Theorem 3.4 Assume that the premise for a discrete MISO fuzzy system is given by (37).

$$B^*(y) = \overline{B_i(y)}, \quad \text{for OFMT} \quad (37)$$

Approximate reasoning results by OFMT-EDM are obtained as

$$A_1^*(x) = A_{i1}(x) \text{ and } A_2^*(x) = A_{i2}(x) \text{ and } \cdots A_j^*(x) = A_{ij}(x) \cdots \text{ and } \cdots A_m^*(x) = A_{im}(x) \quad (38)$$

, satisfies the reductive property by 100%, and fully in accordance with human thinking.

4. Simulation and Results Analysis

4.1 Simulation and Results Analysis in SISO Fuzzy System

Experiments were carried out based on the sample data in [23] for comparative analysis. In the case of SISO fuzzy systems with different input/output dimensions, i.e., input and given preconditions in OFMP-EDM, 5*1-D row vectors, 7*1-D row vectors, and 7*1-D row vectors of input and given preconditions in OFMT-EDM, the experimental results for inference and reductive property estimation by proposed method OAFR for class 1 are shown in Table 5. In the experiment by Matlab, the sign vector is taken as $p(+1, -1)$ form and the threshold is set to $\varepsilon = 0.0025$.

Table 5. Inference results and reductive property of the proposed method

Proposed OFMP-EDM Reasoning result and Reductive property

Case	Premise $A^*(x)$	Reasoning result $B^*(y)$	Reductive property
1	[1, 0.3, 0, 0, 0]	[0, 0, 0, 0.3, 1]	100 %
2	[1, 0.09, 0, 0, 0]	[2e-07, 0, 0, 0, 2e-07, 0.3, 1]	97.00 %
3	[1, 0.548, 0, 0, 0]	[0, 2.3e-07, 2.3e-07, 2.3e-07, 0, 0.3, 1]	96.46 %
4	[0, 0.7, 1, 1, 1]	[0.035, 1, 1, 1, 1, 0.93, 0]	98.58 %
$RPCF_{OFMP}$		$OFMP - EDM - p(+1,-1) form$	98.01 %
Proposed OFMT-EDM Reasoning result and Reductive property			
Case	Premise $B^*(y)$	Reasoning result $A^*(x)$	Reductive property
6	[1, 1, 1, 1, 1, 0.7, 0]	[0, 0.7, 1, 1, 1]	100 %
7	[1, 1, 1, 1, 1, 0.91, 0]	[0, 0.7, 1, 1, 1]	95.80 %
8	[1, 1, 1, 1, 1, 0.45, 0]	[0, 0.7, 1, 1, 1]	95.05 %
9	[0, 0, 0, 0, 0, 0.3, 1]	[1, 0.064, 0, 0, 0]	96.84 %
$RPCF_{OFMT}$		$OFMP - EDM - p(+1,-1) form$	96.92 %

307 The results of optimal fuzzy approximate reasoning by Zadeh's CRI method in the same case are shown
308 in Table 6.

309 **Table 6.** The results of optimal fuzzy approximate reasoning by Zadeh

OFMP-CRI Reasoning result and Reductive property
--

Case	Premise $A^*(x)$	Reasoning result $B^*(y)$	Reductive property
1	[1, 0.3, 0, 0, 0]	[0, 0, 0, 0, 0, 0.3, 1]	100 %
2	[1, 0.09, 0, 0, 0]	[0, 0, 0, 0, 0, 0.3, 1]	97.00 %
3	[1, 0.548, 0, 0, 0]	[0.25, 0.25, 0.25, 0.25, 0.25, 0.55, 1]	82.31 %
4	[0, 0.7, 1, 1, 1]	[1, 1, 1, 1, 1, 1, 1]	81.43 %
$RPCF_{OFMP}$		$OFMP - CRI$	90.18 %
OFMT-CRI Reasoning result and Reductive property			
Case	Premise $B^*(y)$	Reasoning result $A^*(x)$	Reductive property
6	[1, 1, 1, 1, 1, 0.7, 0]	[1, 1, 1, 1, 1]	74 %
7	[1, 1, 1, 1, 1, 0.91, 0]	[1, 1, 1, 1, 1]	78.2 %
8	[1, 1, 1, 1, 1, 0.45, 0]	[1, 1, 1, 1, 1]	69.05 %
9	[0, 0, 0, 0, 0, 0.3, 1]	[1, 1, 1, 1, 1]	26 %
$RPCF_{OFMT}$		$OFMT - CRI$	61.81 %

Comparison and analysis of the above experimental results are as follows.

For the discrete SISO fuzzy system, in the simulation by Matlab, the algorithm computation is completed at iteration $t = 11$. Applying the optimization approach to the Zadeh's CRI method, reductive property improved from 88% to 90.18% for OFMP-CRI and from 61.81% to 76% for OFMT-CRI, respectively. And the method in [16] improved the reductive property by the proposed method from 87.64% to 98.01% for FMP and from 88.67% to 97.38% for FMT, respectively. According to the simulation results, for OFMP-EDM in the case of SISO fuzzy system, the proposed method OFMP-EDM improved the reductive property

317 by 7.83% compared to Zadeh's CRI method [1], and 35.11% for OFMT-EDM significantly. The overall
318 reductive property is that the proposed method OFAR improves the CRI method by 21.38%. Considering
319 the experimental results in detail, the reductive property of OFMP and OFMT of the CRI method is very
320 different. Thus, the reductive property of FMP is high, but the reductive property of FMT is relatively low.
321 The reason is that the CIR method does not properly apply the axiom that FMT is opposite to FMP in
322 general, thus increasing the loss of information by nonlinear operators in matrix operations. As shown in
323 Case 9 of Table 3 above, it can be seen that for the four given premises, the results of each reasoning are
324 all $[1, 1, 1, 1, 1]$, which are not at all compatible with human thinking, as a result of the nonlinear operator.
325 Specifically, as shown in Table 3, the reductive property of OFMT-CRI is 26% for Case 9 and hardly
326 corresponds to human thinking. However, the proposed method OFAR performs approximate reasoning by
327 correctly implementing the dual-order opposite theorem in each procedure of the algorithm shown above,
328 and uses linear operators and normalization operations to reduce the loss of information by max and min
329 nonlinear operators, thus having reductive property over 96% for OFMP-EDM and over 95% for OFMT-
330 EDM for four given premises. On the other hand, by the introduction of distance measure in SISO fuzzy
331 system, reductive property reached 87.64% in FMP-DM, 92.98% in FMP-DM, and 90.31% in total, and
332 significantly improved the reasoning results of previous CRI, AARS, TIP and QIP methods [1,23-25],
333 respectively. The AARS, TIP, and QIP methods [23-25] do not improve the reasoning results better than the
334 CRI method even by applying the optimization approach. The reason is that these methods have higher
335 information loss due to nonlinear operation compared to the CRI method, respectively.

336 In this paper, we improved the reductive property of FMT-DM and FMP-DM by 8% and 3%, respectively,
337 and in total by 7.07% compared to the previous DMM method [16]. This shows that the optimization by
338 introducing an objective function is effective and valid for fuzzy approximation reasoning. Through the
339 experimental results, we conclude that the proposed method OAFR is better in accordance with human

340 thinking compared to the previous methods [1,16,23-25] in SISO fuzzy systems.

341 **4.2 Experiment and results analysis in Discrete MISO fuzzy system**

342 The reductive property criteria of the discrete MISO fuzzy system discussed in this paper is shown in
343 Table 7.

344 **Table 7.** New reductive property criteria for discrete MISO fuzzy systems

FMP	$R_i : \text{if } x_1 \text{ is } A_{i1} \cdots x_j \text{ is } A_{ij} \cdots x_m \text{ is } A_{im} \cdots \text{then } y_i \text{ is } B_i$
Case 1	$RPCF_{FMP}^{MISO} = (1 - \sum_{k=1}^r b_{ki}^* - b_{ki} / r) \times 100$
Case 2	$RPCF_{FMP}^{MISO} = (1 - \sum_{k=1}^r b_{ki}^* - b_{ki}^2 / r) \times 100$
Case 3	$RPCF_{FMP}^{MISO} = (1 - \sum_{k=1}^r b_{ki}^* - b_{ki}^{1/2} / r) \times 100$
Case 4	$RPCF_{FMP}^{MISO} = (1 - \sum_{k=1}^r b_{ki}^* - (1 - b_{ki}) / r) \times 100$
Case 5	$RPCF_{FMP}^{MISO} = (1 - \sum_{k=1}^r b_{ki}^* - s.t.b_{ki} / r) \times 100$
FMT	$R_i : \text{if } y_i \text{ is } \bar{B}_i \text{ then } x_1 \text{ is } \bar{A}_{i1} \cdots x_j \text{ is } \bar{A}_{ij} \cdots x_m \text{ is } \bar{A}_{im}$
Case 6	$RPCF_{FMT-j}^{MISO} = (1 - \min_{i=1}^n \sum_{k=1}^r a_{kij}^* - (1 - a_{kij}) / r) \times 100$
Case 7	$RPCF_{FMT-j}^{MISO} = (1 - \min_{i=1}^n \sum_{k=1}^r a_{kij}^* - (1 - a_{kij}^2) / r) \times 100$
Case 8	$RPCF_{FMT-j}^{MISO} = (1 - \min_{i=1}^n \sum_{k=1}^r a_{kij}^* - (1 - a_{kij}^{1/2}) / r) \times 100$
Case 9	$RPCF_{FMT-j}^{MISO} = (1 - \min_{i=1}^n \sum_{k=1}^r a_{kij}^* - a_{kij} / r) \times 100$
Case 10	$RPCF_{FMT-j}^{MISO} = (1 - \min_{i=1}^n \sum_{k=1}^r a_{kij}^* - s.t.a_{kij} / r) \times 100$

345 In the experiment by Matlab, the threshold was set to $\varepsilon=0.0025$. Table 8 shows the approximate
346 reasoning results for discrete MISO fuzzy systems.

347 **Table 8.** Conclusions and reductive property of the proposed method for a discrete MISO fuzzy system
348 with two-rule

OFMP-EDM	Reasoning result and the reductive property of OFMP-EDM
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Premise $A_l^*(x), A_2^*(x)$	Reasoning result $B^*(y)$		RPCF
$[1, 0.3, 0, 0, 0],$	$P(+1, 0, -1) form$	$[0, 0, 0, 0.3, 1, 1, 1]$	100(%)
$[0, 0, 0, 0.3, 1]$	$P(+1, -1) form$	$[0, 0, 0, 0.3, 1, 1, 1]$	100(%)
$[1, 0.09, 0, 0, 0],$	$P(+1, 0, -1) form$	$[0, 0, 0, 0.14, 0.84, 0.84, 1]$	94.72(%)
$[0, 0, 0, 0.09, 1]$	$P(+1, -1) form$	$[0, 0, 0, 0.14, 0.84, 0.84, 1]$	94.72(%)
$[1, 0.55, 0, 0, 0],$	$P(+1, 0, -1) form$	$[0, 0, 0, 0.3, 1, 1, 1]$	96.46(%)
$[0, 0, 0, 0.55, 1]$	$P(+1, -1) form$	$[0, 0, 0, 0.3, 1, 1, 1]$	96.46(%)
$[0, 0.7, 1, 1, 1],$	$P(+1, 0, -1) form$	$[1, 1, 1, 0.7, 0, 0, 0]$	100(%)
$[1, 1, 1, 0.7, 0]$	$P(+1, -1) form$	$[1, 1, 1, 0.7, 0, 0, 0]$	98.57(%)
RPCF	OFMP-EDM $P(+1, 0, -1) form$		97.80(%)
	OFMP-EDM $P(+1, -1) form$		97.44(%)
OFMT-EDM	Reasoning result and the reductive property of OFMT-EDM		
Premise $B^*(y)$	Reasoning result $A^*(x)$		RPCF
$[1, 1, 1, 0.7, 0, 0, 0]$	$P(+1, 0, -1) form$	$[0, 0.7, 1, 1, 1]$ and $[1, 1, 1, 0.7, 0]$	100.0(%)
	$P(+1, -1) form$	$[0, 0.7, 1, 1, 1]$ and $[1, 1, 1, 0.7, 0]$	100.0(%)
$[1, 1, 1, 0.91, 0, 0, 0]$	$P(+1, 0, -1) form$	$[0, 0.64, 0.91, 1, 0.91]$ and $[1, 1, 1, 0.79, 0]$	95.80(%)
	$P(+1, -1) form$	$[0, 0.7, 1, 1, 1]$ and $[1, 1, 1, 0.7, 0]$	95.80(%)
$[1, 1, 1, 0.45, 0, 0, 0]$	$P(+1, 0, -1) form$	$[0, 0.7, 1, 0.89, 1]$ and $[1, 1, 1, 0.59, 0]$	93.26(%)
	$P(+1, -1) form$	$[0, 0.7, 1, 0.78, 1]$ and $[1, 1, 1, 0.48, 0]$	93.26(%)
$[0, 0, 0, 0.3, 1, 1, 1]$	$P(+1, 0, -1) form$	$[0.55, 0.16, 0, 0, 1]$ and	100(%)

		[0,0,0,0.11,1]	
	$P(+1,-1)form$	[0.55,0.16,0,0,1]	and
		[0,0,0,0.11,1]	98(%)
	OFMT-EDM	$P(+1,0,-1)form$	97.26(%)
RPCF	OFMT-EDM	$P(+1,-1)form$	96.76(%)

Analyzing the experimental results for the discrete MISO fuzzy system is as follows. In the Matlab experiment on OFMP-EDM $P(+1,0,-1)form$, the algorithm computation was completed when the of iterations of optimal fuzzy approximate reasoning is $t=1$ for a given fuzzy premise vector $A_1^*(x) = A_{11}(x), A_2^*(x) = A_{12}(x), i = 1,2$. When the given premise is $A_1^*(x) = very A_{11}(x), A_2^*(x) = very A_{12}(x), i = 1,2$, the algorithm computation is completed at $t = 13$. And when $A_1^*(x) = more or less A_{11}(x), A_2^*(x) = more or less A_{12}(x), i = 1,2$ is $t=12$, and when the given premise is $A_1^*(x) = not A_{11}(x), A_2^*(x) = not A_{12}(x), i = 1,2$ the algorithm computation is completed at $t=2$. In the Matlab experiment for the proposed method OFMP-EDM $P(+1,-1)form$ in the discrete MISO fuzzy system, algorithm terminates at $t=1, t=10, t=11$ and $t=2$ for the same case, respectively. Through the experiment by the proposed method, the overall reductive property of the optimal fuzzy approximation trend results for the four given premises reached over 97%, improved over 7% of the reductive property by [16] (90%), and over 20% of the CRI method. According to a previous study [16], the methods of TIP, QIP, AARS, etc. are not compared with those of the method in this paper because they are less reductive than the CRI method. According to Matlab experiment, in the case of discrete MISO fuzzy system the reductive property of the proposed method was slightly less than the case of SISO fuzzy system. This is because increasing the number of rules leads to loss of information when synthesizing partial reasoning results. However, our proposed method was verified by experiments with significantly higher reductive property compared to the previous methods.

Comprehensively the reductive property of our proposed method was improved than the previous one as

367 over 1.08 times for OFMP, 1.56 times for OFMT, and averagely 1.32 times for OFAR, respectively.

368 **5. Conclusions**

369 In this paper, we proposed new optimal fuzzy approximate reasoning algorithm based on KSI Principle,
370 which can open a new research direction in the field of fuzzy inference. This paper showed that the
371 theoretical and the experimental results are consistent. The experimental comparison between the proposed
372 method OFAR and the previous fuzzy inference methods obtained the conclusion that OFAR is more
373 reductive than the previous methods and is better suited to human thinking. In the future, we will apply the
374 idea and method proposed in this paper to the robustness problem of fuzzy approximate reasoning to discuss
375 its effectiveness and also propose real-world applications.

376

377 **Conflict of interest**

378 The authors declare that they have no conflict of interest.

379 **Authors contribution statement**

380 Son-Il Kwak, Yon-Ju Jang, Chol-Jun Hwang and Ok-Sim Ri wrote the main manuscript text.

381 Son-Il Kwak and Hyon-U Yu proposed propose a new optimal approximate reasoning method based
382 on KSI principle. Our new method is based on obtaining fuzzy reasoning results based on moving
383 difference obtained from a given fuzzy premise and fuzzy antecedents, obtaining new fuzzy premises
384 from the moving difference of this reasoning results and the consequent fuzzy set, and then iteratively
385 obtaining the new reasoning results different from the previous reasoning results by fuzzy moving
386 difference obtained from these fuzzy premises and the antecedent fuzzy set until the objective function
387 is satisfied.

388 Chol-Jun Hwang and Yon-Ju Jang proposed new optimal fuzzy approximate reasoning algorithm
389 based on KSI Principle, which can open a new research direction in the field of fuzzy inference. Hyok-

390 Gi Chae prepared tables 1-4. Ok-Sim Ri prepared tables 5-8.

391 All authors reviewed the manuscript.

392

393 **Funding Statement**

394 No funding was received to assist with the preparation of this manuscript.

395 **Ethical and informed consent for data used**

396 This article does not contain any studies on human participants or animals performed by any of the
397 authors.

398 **Data availability and access**

399 The research period is from March 2021 to September 2023.

400 Zadeh proposed fuzzy set theory, proposed fuzzy modus ponens and fuzzy modus tollens, and
401 discovered fuzzy inference method with the identifier [https://doi.org/10.1016/0020-0255\(75\)90036-5](https://doi.org/10.1016/0020-0255(75)90036-5)

402 Quintuple Implication Principle (QIP) with the identifier [https://doi.org/10.1016/S0165-](https://doi.org/10.1016/S0165-0114(99)80008-6)
403 0114(99)80008-6

404 New idea of fuzzy inference based on optimization techniques with the identifier
405 <https://doi.org/10.1016/j.camwa.2006.08.021>

406 The topic of robustness (Robustness) with the identifier <https://doi.org/10.1109/91.963760>

407 The idea of the optimal fuzzy inference method based on Zadeh's CRI with the identifier
408 <https://dl.acm.org/doi/10.5555/1714810.1715326>

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